Lesson 11 Teacher's Key: Wave After Wave

001 Game Creator (Windows only) - Instructor Reference Only

VIDEO GAME CREATION

What This Lesson Teaches

Splitting the existing ResetGame Custom Event into two purpose-specific Custom Events (ResetGame for full resets, SpawnWave for just reviving enemies) - directly extending last week's introduction of Custom Events as this engine's reusable-subroutine feature.

Level 1 - Full Worked Solution

- SpawnWave contains exactly the revive/reposition actions that used to live inside ResetGame - moved, not duplicated.
- ResetGame is now a thin wrapper: reset all Variables including the new wave, then Call Custom Event(SpawnWave) to handle the actual enemy revival.
- The 'Collided with actor' trigger's nested If is where the real branching logic lives - a genuinely two-part decision this week, verified against the same wiki-confirmed Custom Event mechanism used last week.

Why This Structure

- This is the clearest payoff yet of introducing Custom Events last week: because the revive/reposition logic was already isolated into its own callable block, splitting it out further and calling it from a second location (the win-check's Else branch) required no new mechanism, just a second Call Custom Event action.
- Keeping the win Message Box reserved for the true, final win (not wave transitions) continues the same discipline from Lesson 6, applied consistently across every lesson since.

Level 2 / Level 3 Talking Points

- Ask students to name every place Call Custom Event(SpawnWave) now appears (ResetGame, and the win-check's Else branch) - two call sites, one Custom Event.
- This is a good lesson to revisit the Lesson 5 design-time-placement decision one more time: because there are only ever exactly 5 named enemy copies (never created or destroyed at runtime, only revived and repositioned), 'a new wave' in this engine literally means the same 5 actors coming back to life - conceptually identical to every other engine's fresh spawn, just implemented differently.

Common Student Mistakes

- Checking only 'enemies_remaining <= 0' without the nested wave check.

- Calling SpawnWave inside the win branch by mistake, causing enemies to revive right as the game is won.
- Forgetting wave=1 inside ResetGame, leaving it at whatever value it had before a restart.